OpenDRIVE® 1.8.1, Static Road Network Description, 2024-11-21

12.1 Introduction to junctions

Junctions enable the connection of more than two roads.

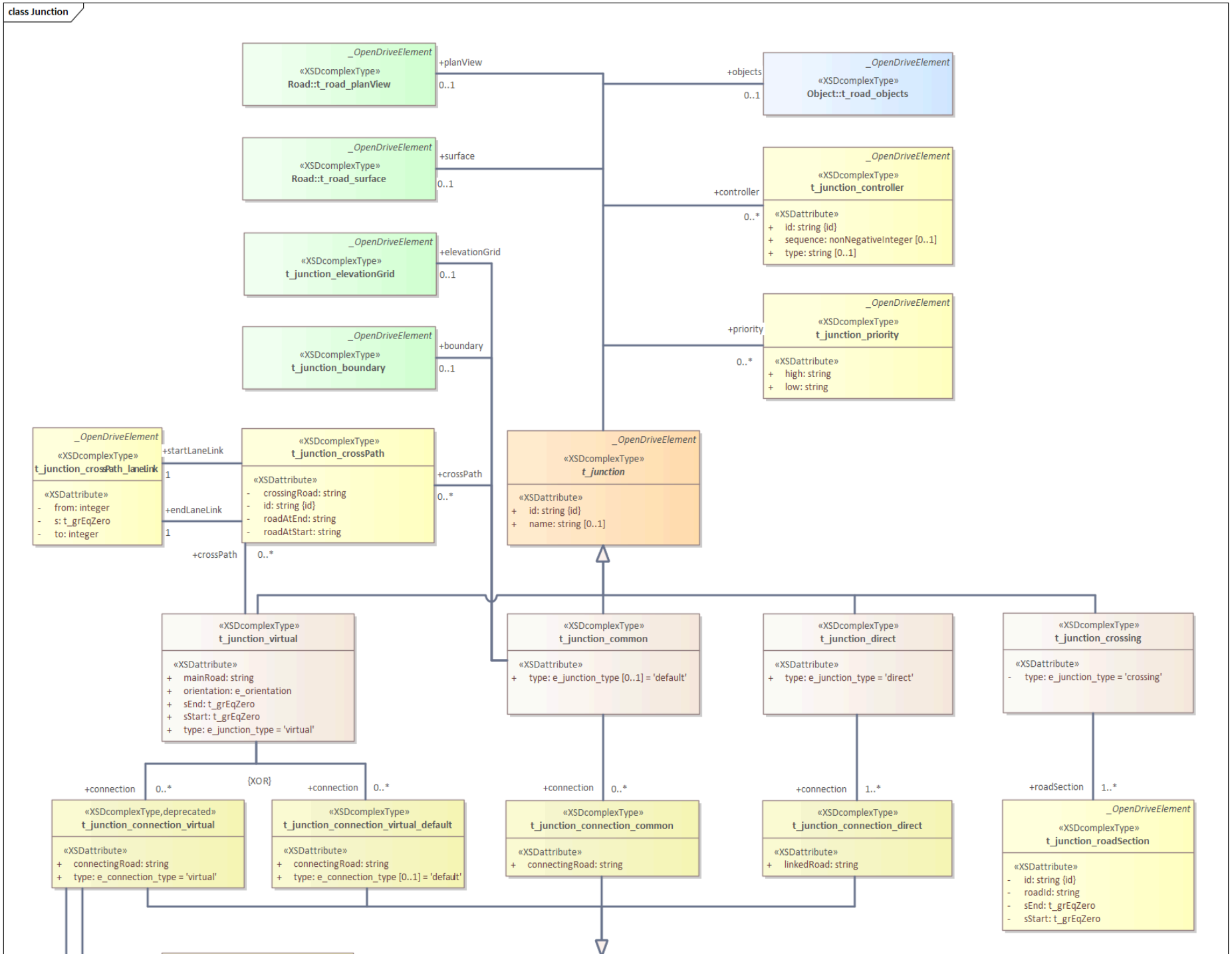
Four types of junctions exist:

- Common junctions are junctions with drivable lanes that can overlap and where traffic can cross.
- Direct junctions are junctions where traffic can change roads but cannot cross other traffic.
- Virtual junctions are junctions where the main road is not interrupted.
- Crossings are junctions where traffic cannot change the roads.

Table 54. Usage of the different types of junctions

Use case	Overlapping lanes	Crossing traffic	Changing roads	Junction type	Alternative junction type
ordinary junctions	yes	yes	yes	common	n/a
junctions with traffic lights	yes	yes	yes	common	n/a
entries and exits	no	no	yes	direct	common (not recommended)
	yes	no	yes	if constant elevation: direct, otherwise: common	common
	yes	yes	yes	common	n/a (direct not possible)
driveways to parking lots	yes	yes	yes	if constant elevation: virtual, otherwise: common	common

Use case	Overlapping lanes	Crossing traffic	Changing roads	Junction type	Alternative junction type
driveways to residential estates	yes	yes	yes	if constant elevation: virtual, otherwise: common	common
slip lanes	no	no	yes	combination of one common and many direct	common
	yes	no	yes	if constant elevation: combination of one common and many direct, otherwise: many common	common
railway crossing	yes	yes	no	crossing	common


class Junction
_OpenDriveElement
«XSDcomplexType»
Road::t_road_planView
+planView
0..1
_OpenDriveElement
«XSDcomplexType»
Road::t_road_surface
+surface
0..1
_OpenDriveElement
«XSDcomplexType»
t_junction_elevationGrid
+elevationGrid
0..1
_OpenDriveElement
«XSDcomplexType»
t_junction_boundary
+boundary
0..1
+objects
0..1
_OpenDriveElement
«XSDcomplexType»
Object::t_road_objects
+controller
0..*
_OpenDriveElement
«XSDcomplexType»
t_junction_controller
«XSDattribute»
+ id: string {id}
+ sequence: nonNegativeInteger [0..1]
+ type: string [0..1]
+priority
0..*
_OpenDriveElement
«XSDcomplexType»
t_junction_priority
«XSDattribute»
+ high: string
+ low: string
_OpenDriveElement
«XSDcomplexType»
t_junction_crossPath_laneLink
«XSDattribute»
- from: integer
- s: t_grEqZero
- to: integer
+startLaneLink
1
+endLaneLink
1
«XSDcomplexType»
t_junction_crossPath
«XSDattribute»
- crossingRoad: string
- id: string {id}
- roadAtEnd: string
- roadAtStart: string
+crossPath
0..*
_OpenDriveElement
«XSDcomplexType»
t_junction
«XSDattribute»
+ id: string {id}
+ name: string [0..1]
+crossPath
0..*
«XSDcomplexType»
t_junction_virtual
«XSDattribute»
+ mainRoad: string
+ orientation: e_orientation
+ sEnd: t_grEqZero
+ sStart: t_grEqZero
+ type: e_junction_type = 'virtual'
«XSDcomplexType»
t_junction_common
«XSDattribute»
+ type: e_junction_type [0..1] = 'default'
«XSDcomplexType»
t_junction_direct
«XSDattribute»
+ type: e_junction_type = 'direct'
«XSDcomplexType»
t_junction_crossing
«XSDattribute»
- type: e_junction_type = 'crossing'
+connection
0..*
{XOR}
+connection
0..*
+connection
0..*
+connection
1..*
+roadSection
1..*
«XSDcomplexType,deprecated»
t_junction_connection_virtual
«XSDattribute»
+ connectingRoad: string
+ type: e_connection_type = 'virtual'
«XSDcomplexType»
t_junction_connection_virtual_default
«XSDattribute»
+ connectingRoad: string
+ type: e_connection_type [0..1] = 'default'
«XSDcomplexType»
t_junction_connection_common
«XSDattribute»
+ connectingRoad: string
«XSDcomplexType»
t_junction_connection_direct
«XSDattribute»
+ linkedRoad: string
_OpenDriveElement
«XSDcomplexType»
t_junction_roadSection
«XSDattribute»
- id: string {id}
- roadId: string
- sEnd: t_grEqZero
- sStart: t_grEqZero

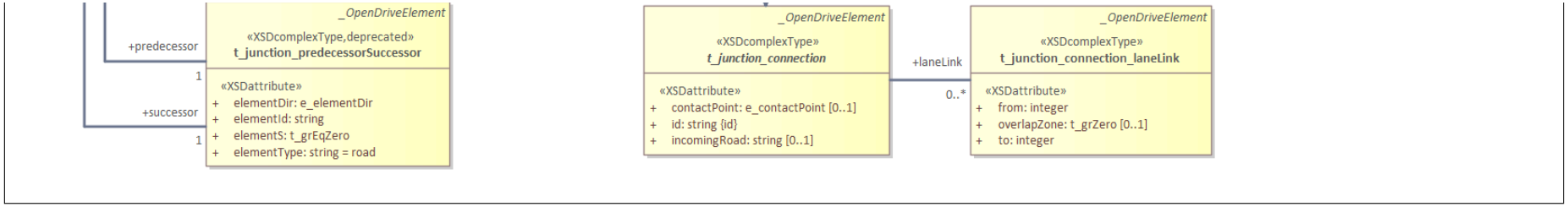


Figure 82. UML class diagram of the Junction class

Figure 82 shows the UML class diagram of the ASAM OpenDRIVE Junction class.

Rules

The following rules apply to junctions:

- asam.net:xodr:1.4.0:junctions.no_overlap: No junctions of any type shall overlap each other.
- asam.net:xodr:1.7.0:junctions.type_direct_no_conn_road: The `<connection>` element of a junction of @type="direct" shall not have the @connectingRoad attribute.
- asam.net:xodr:1.7.0:junctions.type_default_no_linked_road: The `<connection>` element of a junction of @type="default" or @type="virtual" shall not have the @linkedRoad attribute.